Marketing

Description

Problem statement

A restaurant consolidator is looking to revamp the B2C portal using intelligent automation tech. This requires a different matrix to identify the star restaurants and generate recommendations. To make sure an effective model can be achieved, it is important to understand the behavior of the data in hand.

There are 2 datasets, data and Country-Code. Dataset data has 19 attributes and Country-Code has two attributes.

You are required to determine certain matrices to identify the star restaurants and generate recommendations.

Importing, Understanding, and Inspecting Data:

Perform preliminary data inspection and report the findings as the structure of the data, missing values, duplicates, etc.

Based on the findings from the previous questions, identify duplicates and remove them

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

[3]:
```
df=pd.read_excel("Data.xlsx")
```

[5]:
```
df.head()
```

[5]:

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	Average Cost for two	Currency	Has Table booking	Has Online delivery	Price range	A
0	7402935	Skye	94	Jakarta	Menara BCA, Lantai 56, Jl. MH. Thamrin, Thamri...	Grand Indonesia Mall, Thamrin	Grand Indonesia Mall, Thamrin, Jakarta	106.821999	-6.196778	Italian, Continental	800000	Indonesian Rupiah(IDR)	No	No	3	
1	7410290	Satoo - Hotel Shangri-La	94	Jakarta	Hotel Shangri-La, Jl. Jend. Sudirman	Hotel Shangri-La, Sudirman	Hotel Shangri-La, Sudirman, Jakarta	106.818961	-6.203292	Asian, Indonesian, Western	800000	Indonesian Rupiah(IDR)	No	No	3	
2	7420899	Sushi Masa	94	Jakarta	Jl. Tuna Raya No. 5, Penjaringan	Penjaringan	Penjaringan, Jakarta	106.800144	-6.101298	Sushi, Japanese	500000	Indonesian Rupiah(IDR)	No	No	3	
3	7421967	3 Wise Monkeys	94	Jakarta	Jl. Suryo No. 26, Senopati, Jakarta	Senopati	Senopati, Jakarta	106.813400	-6.235241	Japanese	450000	Indonesian Rupiah(IDR)	No	No	3	
					Gedung PIC, Jl.											

[7]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9550 entries, 0 to 9549
Data columns (total 19 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Restaurant ID         9550 non-null   int64
 1   Restaurant Name       9550 non-null   object
 2   Country Code          9550 non-null   int64
 3   City                  9550 non-null   object
 4   Address               9550 non-null   object
 5   Locality              9550 non-null   object
 6   Locality Verbose      9550 non-null   object
 7   Longitude             9550 non-null   float64
 8   Latitude              9550 non-null   float64
 9   Cuisines              9541 non-null   object
 10  Average Cost for two  9550 non-null   int64
 11  Currency              9550 non-null   object
 12  Has Table booking     9550 non-null   object
 13  Has Online delivery   9550 non-null   object
 14  Price range           9550 non-null   int64
 15  Aggregate rating      9550 non-null   float64
 16  Rating color          9550 non-null   object
 17  Rating text           9550 non-null   object
 18  Votes                 9550 non-null   int64
dtypes: float64(3), int64(5), object(11)
memory usage: 1.4+ MB
```

[9]:
```
df.describe()
```

[9]:

	Restaurant ID	Country Code	Longitude	Latitude	Average Cost for two	Price range	Aggregate rating	Votes
count	9.550000e+03	9550.000000	9550.000000	9550.000000	9550.000000	9550.000000	9550.000000	9550.000000
mean	9.052064e+06	18.367435	64.125697	25.854675	1199.252565	1.804712	2.666220	156.845654
std	8.791506e+06	56.753239	41.469140	11.008474	16122.026663	0.905574	1.516386	430.146060
min	5.300000e+01	1.000000	-157.948486	-41.330428	0.000000	1.000000	0.000000	0.000000
25%	3.019902e+05	1.000000	77.081380	28.478745	250.000000	1.000000	2.500000	5.000000
50%	6.004248e+06	1.000000	77.191964	28.570470	400.000000	2.000000	3.200000	31.000000
75%	1.835229e+07	1.000000	77.282012	28.642760	700.000000	2.000000	3.700000	131.000000
max	1.850065e+07	216.000000	174.832089	55.976980	800000.000000	4.000000	4.900000	10934.000000

[11]:
```
df.columns
```

[11]:
```
Index(['Restaurant ID', 'Restaurant Name', 'Country Code', 'City', 'Address',
       'Locality', 'Locality Verbose', 'Longitude', 'Latitude', 'Cuisines',
       'Average Cost for two', 'Currency', 'Has Table booking',
       'Has Online delivery', 'Price range', 'Aggregate rating',
       'Rating color', 'Rating text', 'Votes'],
      dtype='object')
```

[13]:
```
df.shape
```

[13]:
```
(9550, 19)
```

[15]:
```
df.isnull().sum()
```

[15]:
```
Restaurant ID           0
Restaurant Name         0
Country Code            0
City                    0
Address                 0
Locality                0
Locality Verbose        0
Longitude               0
Latitude                0
Cuisines                9
Average Cost for two    0
Currency                0
Has Table booking       0
Has Online delivery     0
Price range             0
Aggregate rating        0
Rating color            0
Rating text             0
Votes                   0
dtype: int64
```

Explore the geographical distribution of the restaurants and identify the cities with the maximum and minimum number of restaurants

```
[29]: df1.Country.value_counts()
```

```
[29]: Country
      India             8651
      United States      434
      United Kingdom      80
      South Africa        60
      UAE                 60
      Brazil              60
      New Zealand         40
      Turkey              34
      Australia           24
      Phillipines         22
      Indonesia           21
      Sri Lanka           20
      Qatar               20
      Singapore           20
      Canada               4
      Name: count, dtype: int64
```

```
[35]: df1.City.value_counts()
```

```
[35]: City
      New Delhi    5473
      Gurgaon      1118
      Noida        1080
      Faridabad     251
      Ghaziabad      25
                   ...
      Lincoln         1
      Monroe          1
      Potrero         1
      Weirton         1
      Miller          1
      Name: count, Length: 141, dtype: int64
```

Restaurant franchising is a thriving venture. So, it is very important to explore the franchise with most national presence

```
[115]: RestaurantName_Counts=df1.groupby('Restaurant Name')['Country'].nunique()

[117]: most_national_presence=RestaurantName_Counts.idxmax()

[119]: most_national_presence1=RestaurantName_Counts.max()

[125]: print (most_national_presence)

       Hard Rock Cafe

[131]: print(f"The franchise with the most national presence is '{most_national_presence}' in {most_national_presence1} Country.")

       The franchise with the most national presence is 'Hard Rock Cafe' in 3 Country.
```

Find out the ratio between restaurants that allow table booking vs. those that do not allow table booking

```
[137]: allow_booking_count = df1[df1['Has Table booking'] == 'Yes'].shape[0]

[139]: no_booking_count = df1[df1['Has Table booking'] == 'No'].shape[0]

[141]: print(allow_booking_count)

       1158

[143]: print(no_booking_count)

       8392

[145]: if no_booking_count != 0:
           ratio = allow_booking_count / no_booking_count
           print(f"Ratio of restaurants allowing booking to those not allowing booking: {ratio:.2f}")
       else:
           print("No restaurants found that do not allow table booking.")

       Ratio of restaurants allowing booking to those not allowing booking: 0.14
```

Find out the percentage of restaurants providing online delivery

```
[151]: total_restaurants = df1.shape[0]

[153]: delivery_count = df1[df1['Has Online delivery'] == 'Yes'].shape[0]

[155]: percentage_delivery = (delivery_count / total_restaurants) * 100

[157]:
print(f"Percentage of restaurants providing online delivery: {percentage_delivery:.2f}%")

Percentage of restaurants providing online delivery: 25.66%
```

Calculate the difference in number of votes for the restaurants that deliver and the restaurants that do not deliver

```
159]: vote_sums = df1.groupby('Has Online delivery')['Votes'].sum()

175]: difference = vote_sums['Yes'] - vote_sums['No']

177]: print(f"The difference in votes between restaurants that deliver and those that do not deliver is: {difference}")

The difference in votes between restaurants that deliver and those that do not deliver is: -462048
```

What are the top 10 cuisines served across cities? What is the maximum and minimum number of cuisines that a restaurant serves? Also, which is the most served cuisine across the restaurant for each city?

```
[181]: top_cuisines = df1['Cuisines'].value_counts().head(10)

[183]: print("Top 10 Cuisines Across Cities:")
print(top_cuisines)

Top 10 Cuisines Across Cities:
Cuisines
North Indian                      936
North Indian, Chinese             511
Chinese                           354
Fast Food                         354
North Indian, Mughlai             334
Cafe                              299
Bakery                            218
North Indian, Mughlai, Chinese    197
Bakery, Desserts                  170
Street Food                       149
Name: count, dtype: int64
```

What is the distribution cost across the restaurants?

```
[195]: print("Cost Distribution Summary:")
print(df1['Average Cost for two'].describe())

Cost Distribution Summary:
count      9550.000000
mean       1199.252565
std       16122.026663
min           0.000000
25%         250.000000
50%         400.000000
75%         700.000000
max      800000.000000
Name: Average Cost for two, dtype: float64
```

How ratings are distributed among the various factors?

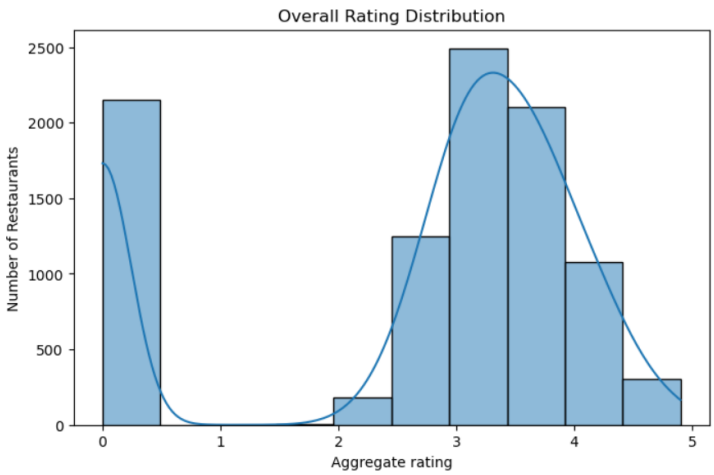
```
[207]: df1.groupby(['Aggregate rating','Rating color','Rating text']).size().reset_index().rename(columns={0:'Rating Count'})
```

[207]:

	Aggregate rating	Rating color	Rating text	Rating Count
0	0.0	White	Not rated	2148
1	1.8	Red	Poor	1
2	1.9	Red	Poor	2
3	2.0	Red	Poor	7
4	2.1	Red	Poor	15
5	2.2	Red	Poor	27
6	2.3	Red	Poor	47
7	2.4	Red	Poor	87
8	2.5	Orange	Average	110
9	2.6	Orange	Average	191
10	2.7	Orange	Average	250
11	2.8	Orange	Average	315
12	2.9	Orange	Average	381
13	3.0	Orange	Average	468
14	3.1	Orange	Average	519
15	3.2	Orange	Average	522
16	3.3	Orange	Average	483

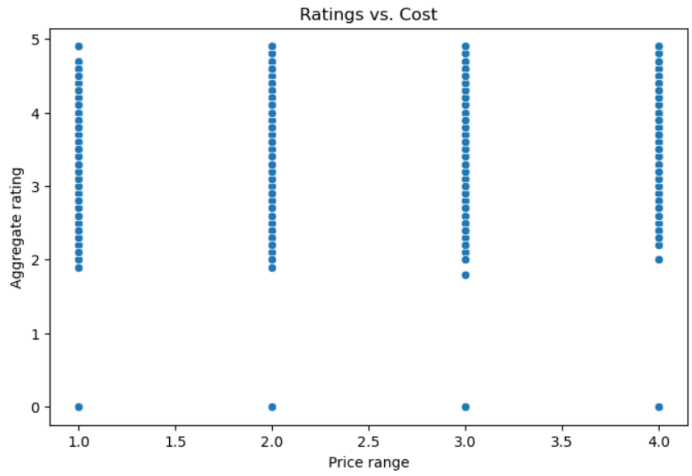
Explain the factors in the data that may have an effect on ratings. For example, number of cuisines, cost, delivery option, etc.

```
[213]: plt.figure(figsize=(8, 5))
sns.histplot(df1['Aggregate rating'], bins=10, kde=True)
plt.title('Overall Rating Distribution')
plt.xlabel('Aggregate rating')
plt.ylabel('Number of Restaurants')
plt.show()
```



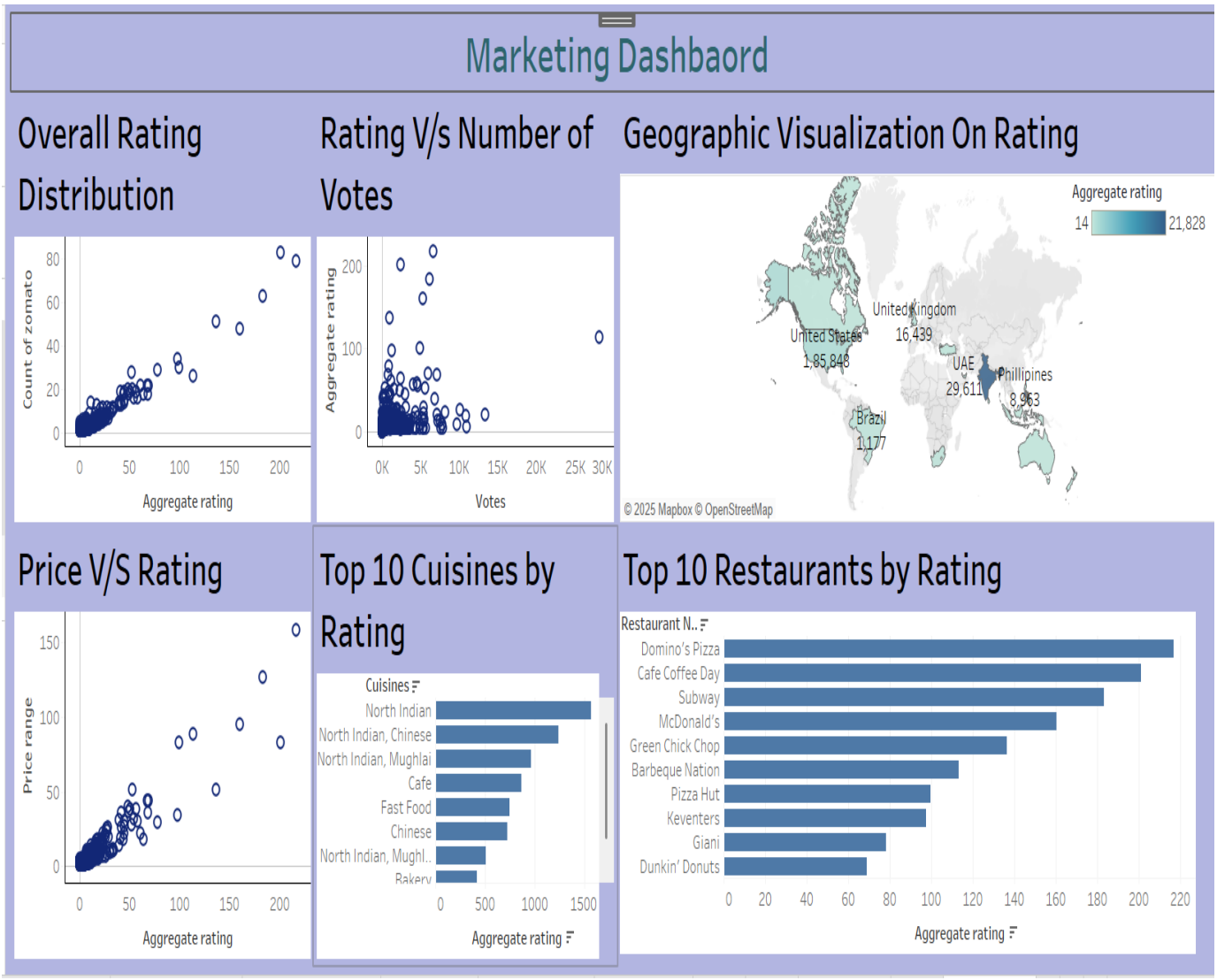
Aggregate rating

[223]:
```
plt.figure(figsize=(8, 5))
sns.scatterplot(x='Price range', y='Aggregate rating', data=df1)
plt.title('Ratings vs. Cost')
plt.xlabel('Price range')
plt.ylabel('Aggregate rating')
plt.show()
```



Price range

[225]:
```
plt.figure(figsize=(6, 4))
sns.boxplot(x='Has Online delivery', y='Aggregate rating', data=df)
plt.title('Ratings by Delivery Option')
plt.xlabel('Has Online delivery')
plt.ylabel('Aggregate rating')
plt.show()
```

Visualize the variables using Tableau to help user explore the data and create a better understanding of the restaurants to identify the ''star'' restaurant

Demonstrate the variables associated with each other and factors to build a dashboard